

On the stability of the Serrin problem

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Abstract

We investigate stability issues concerning the radial symmetry of solutions to Serrin’s overdetermined problems. In particular, we show that, if u is a solution to $\Delta u = n$ in a smooth domain $\Omega \subset \mathbb{R}^n$, $u = 0$ on $\partial\Omega$ and $|Du|$ is “close” to 1 on $\partial\Omega$, then Ω is “close” to the union of a certain number of disjoint unitary balls.

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1. Introduction

In a remarkable paper [11] Serrin establishes the symmetry of the solution to a wide class of uniformly elliptic overdetermined problems. The statement for the Poisson problem can be summarized as follows:

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If $u \in C^2(\bar{\Omega})$ is a solution to the following problem

$$\begin{cases} \Delta u = n & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \\ \frac{\partial u}{\partial \nu_x} = 1 & \text{on } \partial\Omega, \end{cases}$$

where Ω is a bounded, smooth, connected, open set, and ν_x is its unit outer normal, then Ω is a unitary ball and $u = \frac{|x|^2 - 1}{2}$ up to a translation.

In the proof he adapted a technique (introduced by Aleksandrov) known as “moving planes” in conjunction with a smart refinement of the maximum principle, an overall combination which soon had to become a standard approach to prove symmetry of solutions even in different context [2,4,7,10]. Right after Serrin’s work, Weinberger [15] found a one page long, very elegant alternative proof by simply applying the maximum principle to a suitable auxiliary function. The payoff is a proof strongly relying on the linearity of the Laplace equation and not easily generalizable to more general operators.

Recently yet another proof was given in [3] where the authors observed that the solutions to Poisson problems are subject to an integral inequality and the equality sign, achieved in case of the overdetermined boundary conditions given above, enforces the solution to be radially symmetric. The result is carried out by plugging in Newtonian inequalities (therefore simple algebraic inequalities) for the eigenvalues of the Hessian matrix of u together with the Pohožaev identity (see [8]). Such a method establishes a way for shortening the proof in a rather elementary manner and, at the same time, it also works unchanged for the class of nonuniformly elliptic Hessian operators. Last but not least, being the proof based on integral inequalities, the result is obtained by arguing in a “global” manner, avoiding an extensive use of the maximum principle. This aspect suggested us the subject of the present paper: stability issues. Somehow it seems reasonable to investigate whether under small “perturbations” of the boundary conditions the set Ω remains “close” to the unitary ball. In particular we establish the following result.

Theorem 1. *Let Ω be a bounded, smooth, open set in $\mathbb{R}^n$ and let $u \in C^2(\bar{\Omega})$ be a solution to the Laplace equation*

$$\Delta u = n \quad \text{in } \Omega \tag{1}$$

satisfying the following boundary conditions

$$u = 0 \quad \text{on } \partial\Omega, \tag{2}$$

$$|Du| \leq 1 + \delta \quad \text{on } \partial\Omega, \tag{3}$$

$$\| |Du| - 1 \|_{L^1(\partial\Omega)} \leq \delta |\partial\Omega| \tag{4}$$

for some $\delta > 0$.

There exist two positive constants C and β and a finite number of mutually disjoint balls $B^1, \dots, B^m$ such that

$$||B^i| - \omega_n| \leq C\delta^\beta \quad \text{for } 1 \leq i \leq m, \quad (5)$$

$$\left\| u - \frac{|x - x_i|^2 - 1}{2} \right\|_{C^1(\Omega \cap B^i)} \leq C\delta^\beta \quad \text{for } 1 \leq i \leq m, \quad (6)$$

and

$$\left| \Omega \triangle \bigcup_{i=1}^m B^i \right| \leq C\delta^\beta, \quad (7)$$

ω_n and x_i being respectively the measure of the unitary ball and the center of the ball B^i . The constant $\beta > 0$ only depends on n , while the constant C also depends on the diameter of Ω .

Stability of the Serrin problem with the right-hand side of (1) replaced by $f(u)$ was treated in [1] by adapting the moving planes techniques. The hypotheses were slightly different: the set Ω was supposed connected and instead of the boundary conditions (3)–(4) the authors considered

$$\| |Du| - 1 \|_{C^1(\partial\Omega)} \leq \delta.$$

They proved that, under such conditions, there exists two balls $B_r \subset \Omega \subset B_R$ of radii r and R , respectively, such that

$$R - r \leq C |\log \delta|^{-1/n},$$

the constant C depending on n , on the diameter of Ω and, what is more important, on the a priori estimate of the $C^{2,\alpha}$ regularity of Ω .

In the case of $C^{2,\alpha}$ regularity of Ω we establish the following result.

Theorem 2. *Let $\Omega \in C^{2,\alpha}$ be a bounded, connected, open set in $\mathbb{R}^n$ and let u be a solution to problem (1)–(2) satisfying the condition*

$$||Du| - 1| \leq \delta \quad \text{on } \partial\Omega.$$

Then, there exist a positive constant C , depending on n , on the diameter of Ω and on the regularity of Ω , and a constant μ , depending only on n , such that

$$R_{out} - R_{in} \leq C\delta^\mu, \quad (8)$$

where R_{out} and R_{in} are, respectively, the inradius and the circumradius of Ω . Moreover

$$|1 - R_{in}| \leq C\delta^\mu, \quad |1 - R_{out}| \leq C\delta^\mu.$$

We briefly discuss the assumptions of Theorem 1. First of all it may seem slightly odd that on one hand, according to (4), $|Du|$ is close to 1 in $L^1(\partial\Omega)$ and, on the other hand, through (3) we also give an $L^\infty(\partial\Omega)$ bound to $|Du|$. We could replace both conditions by $\| |Du| - 1 \|_{L^\infty(\partial\Omega)} \leq \delta$ and the proof and the result would remain practically unchanged. On the contrary the result is false if we drop (3). Indeed, when $\Omega = B_1(0) \setminus B_\epsilon(0)$ an explicit calculation shows that, when ϵ is small enough compared to δ , (4) holds true while (6) is clearly false. At a certain extent,

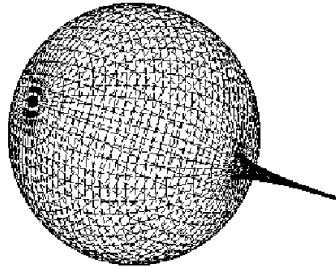


Fig. 1. Our result contemplates the unitary ball with a tiny but arbitrarily long tentacle.

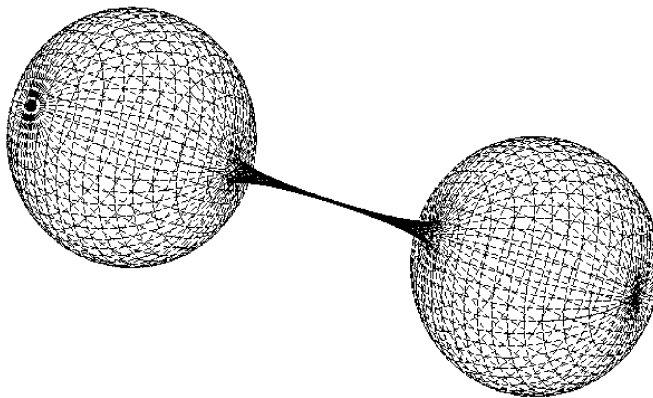


Fig. 2. If the set Ω is made by joining several unitary balls, m (in the statement of Theorem 1) can be greater than one even if Ω is connected.

condition (3) has the advantage to keep “small holes” away from Ω . A different example explains why the constant C in Theorem 1 depends on the diameter of Ω . Let $\Omega = B_{R+2}(0) \setminus B_R(0)$; the conclusions of Theorem 1 are clearly false, however the solution u can be explicitly written down to prove that for any given $\delta > 0$, it is always possible to take R large enough such that the solution u eventually satisfies

$$\| |Du| - 1 \|_{C^\infty(\partial\Omega)} \leq \delta.$$

At last we emphasize that, even assuming Ω connected in Theorem 1, the result holds unchanged and, surprisingly, m can still be greater than 1. Indeed, regardless the value of the constant δ , conditions (3) to (7) hold true if u is the solution to the Poisson problem when Ω is equal to a unitary ball plus an appropriate tiny “tentacle” as in Fig. 1. Inside a feasible tiny tentacle u and $|Du|$ can be taken arbitrarily “small” while in the rest of the set Ω u is just a “perturbation” of the paraboloid $\frac{|x|^2-1}{2}$ (up to a translation).

As a consequence using tiny tentacles it is possible to connect together two or more spheres (see Fig. 2) and the solution u to the Poisson problem still fulfills conditions (3) to (7).

2. Notation and preliminaries

We denote by $A = (a_{ij})$ a matrix in the space S_n of the real symmetric $n \times n$ matrices, and by $\lambda_1, \dots, \lambda_n$ its eigenvalues. Throughout the paper

$$S_1(A) = \lambda_1 + \dots + \lambda_n$$

indicates the trace operator, while we denote by

$$S_2(A) = \sum_{i < j} \lambda_i \lambda_j$$

the sum of all products of two different eigenvalues of A . Note that $S_2(A)$ is just the sum of all 2×2 principal minors of A .

The operator $S_2^{1/2}$ is 1-homogeneous; thus, once set

$$S_2^{ij}(A) = \frac{\partial}{\partial a_{ij}} S_2(A),$$

Euler identity for homogeneous functions gives

$$S_2(A) = \frac{1}{2} S_2^{ij}(A) a_{ij};$$

here and throughout the paper we will adopt the Einstein summation convention for repeated indices.

It is well known that, if $S_1(A) \geq 0$, then

$$\left(\frac{S_1(A)}{n} \right)^2 \geq \frac{S_2(A)}{\binom{n}{2}} \quad (9)$$

and equality holds if and only if $\lambda_1 = \dots = \lambda_n$ (see [6]).

Now let D be a bounded, connected, open set in $\mathbb{R}^n$ with boundary $\partial D \in C^2$, having principal curvatures $\kappa = (\kappa_1, \dots, \kappa_{n-1})$, oriented so that convex sets have positive curvatures. In the following we denote by

$$H_1 = \kappa_1 + \dots + \kappa_{n-1}$$

$(n-1)$ -times the mean curvature of ∂D . Whenever u is a C^2 function and t is a regular value for u , then

$$S_1(D^2 u(x)) = \Delta u(x) = H_1 |Du(x)| + \frac{u_{ij}(x) u_i(x) u_j(x)}{|Du(x)|^2}, \quad (10)$$

where H_1 is $(n - 1)$ -times the mean curvature of the regular level set $\{u = t\}$ passing through x and subscripts stand for partial derivatives. Moreover, the following point-wise identity holds true (see [9,14])

$$H_1 = \frac{S_2^{ij}(D^2u)u_iu_j}{|Du|^3}. \quad (11)$$

Finally, a direct computation yields that $(S_2^{1j}(D^2u), \dots, S_2^{nj}(D^2u))$ is divergence free, i.e.

$$\frac{\partial}{\partial x_i} S_2^{ij} = 0;$$

hence $S_2(D^2u)$ can be written in divergence form

$$S_2(D^2u) = \frac{1}{2} S_2^{ij}(D^2u)u_{ij} = \frac{1}{2} (S_2^{ij}(D^2u)u_j)_i. \quad (12)$$

3. Poisson problem

In this section we prove Theorem 1. For the convenience of the reader, we split the proof in different propositions. For the seek of simplicity we shall assume that u achieves its minimum at 0.

Lemma 3. *Let Ω be a smooth, bounded, open set in $\mathbb{R}^n$ and let $u \in C^2(\bar{\Omega})$ be a solution to the Laplace equation*

$$\Delta u = n \quad \text{in } \Omega \quad (13)$$

satisfying the following boundary conditions

$$u = 0 \quad \text{on } \partial\Omega, \quad (14)$$

$$|Du| \leq 1 + \delta \quad \text{on } \partial\Omega, \quad (15)$$

$$\| |Du| - 1 \|_{L^1(\partial\Omega)} \leq \delta |\partial\Omega|. \quad (16)$$

Then there exists a positive constant C , depending on the diameter d of Ω and on n , such that

$$\int_{\Omega} (-u) \left(\frac{\Delta u}{n} - \frac{S_2(D^2u)}{\binom{n}{2}} \right) \leq C\delta.$$

From now on we will generally denote by C positive constants, depending on d and n , whose values may change from line to line. Moreover, throughout the rest of the paper, we will be free to choose δ as small as needed, since the results are all trivial provided we know how to prove them for all positive δ smaller than some constant δ_0 .

Proof. From Eq. (13) and condition (16) we get

$$(1 - \delta)|\partial\Omega| \leq n|\Omega| \leq (1 + \delta)|\partial\Omega|. \quad (17)$$

Using again Eq. (13) we have

$$\int_{\Omega} (-u) = \frac{1}{n} \int_{\Omega} |Du|^2$$

and thus, by Pohožaev identity (see [12], p. 156), the integral of u over Ω can be calculated as follows

$$\int_{\Omega} (-u) = \frac{1}{n(n+2)} \int_{\partial\Omega} \langle x, \nu \rangle |Du|^2 = \frac{1}{n(n+2)} \int_{\partial\Omega} \langle x, \nu \rangle (|Du|^2 - 1) + \frac{|\Omega|}{n+2}.$$

Moreover, from (15) and (16) we can easily deduce

$$\left| \int_{\partial\Omega} (|Du|^2 - 1) \right| \leq \int_{\partial\Omega} (|Du| + 1) ||Du| - 1| \leq (\delta + 2) \delta |\partial\Omega|,$$

and then we can estimate

$$\frac{|\Omega|}{n+2} - \frac{\delta(\delta+2)d}{n(n+2)} |\partial\Omega| \leq \int_{\Omega} (-u) \leq \frac{|\Omega|}{n+2} + \frac{\delta(\delta+2)d}{n(n+2)} |\partial\Omega|. \quad (18)$$

On the other hand, integrating by parts and using (13), (14) and (10) we have

$$\begin{aligned} \int_{\Omega} |Du|^2 &= \int_{\Omega} |Du|^2 \frac{\Delta u}{n} \\ &= -\frac{2}{n} \int_{\Omega} u_{ij} u_i u_j + \frac{1}{n} \int_{\partial\Omega} |Du|^2 \langle Du, \nu \rangle \\ &= -\frac{2}{n} \int_{\Omega} [\Delta u |Du|^2 - H_1 |Du|^3] + \frac{1}{n} \int_{\partial\Omega} |Du|^3; \end{aligned}$$

then, by (15) and (17),

$$\int_{\Omega} H_1 |Du|^3 \geq \frac{3n^2}{2} \int_{\Omega} (-u) - \frac{n(1+\delta)^3}{2(1-\delta)} |\Omega|. \quad (19)$$

So, by (18), (9), divergence theorem, (12), (11) and (19) we get

$$\begin{aligned} \frac{|\Omega|}{n+2} + \frac{\delta(\delta+2)d}{n(n+2)} |\partial\Omega| &\geq \int_{\Omega} (-u) = \int_{\Omega} (-u) \frac{\Delta u}{n} \\ &\geq \int_{\Omega} (-u) \frac{S_2(D^2u)}{\binom{n}{2}} = \frac{1}{n(n-1)} \int_{\Omega} H_1 |Du|^3 \end{aligned}$$

$$\begin{aligned}
&\geq \frac{3n}{2(n-1)} \int_{\Omega} (-u) - \frac{(\delta+1)^3}{2(n-1)(1-\delta)} |\Omega| \\
&\geq \frac{3n}{2(n-1)} \left(\frac{|\Omega|}{n+2} - \frac{d\delta(\delta+2)}{n(n+2)} |\partial\Omega| \right) - \frac{(\delta+1)^3}{2(n-1)(1-\delta)} |\Omega|,
\end{aligned}$$

that means

$$\int_{\Omega} (-u) \left(\frac{\Delta u}{n} - \frac{S_2(D^2 u)}{\binom{n}{2}} \right) \leq C\delta. \quad \square$$

Remark 4. From (17) and the classical isoperimetric inequality we deduce

$$|\Omega| \geq \omega_n (1-\delta)^n. \quad (20)$$

Moreover, by (18) and (17), we have

$$M = \|u\|_{L^\infty(\Omega)} \geq \frac{1}{n+2} - \frac{\delta(\delta+2)}{(n+2)} \frac{d}{(1-\delta)}.$$

On the other hand, by classical estimates (see, for example, [13]) it holds

$$M \leq \frac{1}{2} \left(\frac{|\Omega|}{\omega_n} \right)^{2/n}.$$

Theorem 5. Under the assumptions of Lemma 3, there exists a positive constant C , depending on d and n , and an open set $E \subset \Omega$ such that any connected component of E is a ball B^i and there exists a $v^i \in C^\infty(\mathbb{R}^n)$, whose graph is a paraboloid (i.e. $D^2 v^i = I$), for which

$$\|u - v^i\|_{C^1(B^i)} \leq C\delta^{\frac{1}{4n+9}}, \quad (21)$$

$$\|u\|_{L^\infty(\Omega \setminus \bigcup_i B^i)} \leq C\delta^{\frac{1}{4n+9}}. \quad (22)$$

Proof. Taking into account Remark 4 let us fix $\epsilon \in (0, M)$ and let us denote by $\Omega_\epsilon = \{u < -\epsilon\}$; by Lemma 3 and (9) we get

$$\int_{\Omega_\epsilon} \left(\frac{\Delta u}{n} - \frac{S_2(D^2 u)}{\binom{n}{2}} \right) \leq C \frac{\delta}{\epsilon}. \quad (23)$$

In the following we shall find a way to choose $\epsilon > 0$ small enough such that $|\Omega_\epsilon| \approx |\Omega|$, but large enough compared to δ so that, from now on, $\frac{\delta}{\epsilon}$ can be considered as a small quantity.

Since $|Du|^2$ is subharmonic in Ω , it attains its maximum on the boundary, hence

$$|Du| \leq 1 + \delta \quad \text{in } \bar{\Omega}. \quad (24)$$

So we can prove that, for every $x \in \Omega_\epsilon$, $\text{dist}(x, \partial\Omega) > \frac{\epsilon}{1+\delta}$. Moreover, from (23) we get the existence of a set $A_\epsilon \subset \Omega_\epsilon$ such that

$$|A_\epsilon| < \left(\frac{\delta}{\epsilon}\right)^{1/2}$$

and

$$\frac{\Delta u}{n} - \frac{S_2(D^2u)}{\binom{n}{2}} \leq C \left(\frac{\delta}{\epsilon}\right)^{1/2} \quad \text{in } \Omega_\epsilon \setminus A_\epsilon. \quad (25)$$

We only have to observe that $|\Omega_\epsilon|$ is not too small. Indeed, since u achieves its minimum at 0 and $\epsilon < M$ it immediately follows that the ball centered at the origin and having radius $\frac{M-\epsilon}{1+\delta}$ is contained in Ω_ϵ ; then $|\Omega_\epsilon| \geq \omega_n \left(\frac{M-\epsilon}{1+\delta}\right)^n$.

Let us observe that by classical estimates (see, for instance [5]) there exists $\bar{C} = \bar{C}(n) > 0$ such that

$$|D^2u(x)| \leq \frac{\bar{C}}{\text{dist}(x, \partial\Omega)^2} \leq \frac{C}{\epsilon^2}, \quad x \in \Omega_\epsilon.$$

From (25) we can easily deduce that D^2u has the following form $u_{ij} = \delta_{ij} + \eta_{ij}$, where $|\eta_{ij}| \leq \eta = C \left(\frac{\delta}{\epsilon}\right)^{1/4}$ in $\Omega_\epsilon \setminus A_\epsilon$. Let $0 < h < \frac{\epsilon}{1+\delta}$ and let us denote by B_h the ball centered at the origin having radius h . Notice that, by (24), and the choice of h , we have $x + B_h \subset \Omega_\epsilon$ for every $x \in \Omega_{2\epsilon}$. Let

$$u_h(x) = \frac{1}{|B_h|} \int_{B_h} u(x-y) dy, \quad x \in \Omega_{2\epsilon}.$$

Then

$$\begin{aligned} \left| \frac{\partial^2 u_h}{\partial x_i^2}(x) - 1 \right| &= \frac{1}{|B_h|} \left| \int_{B_h} \frac{\partial^2 u}{\partial y_i^2}(x-y) dy - |B_h| \right| \\ &\leq \frac{1}{|B_h|} \left(\int_{B_h \cap A_\epsilon} \left| \frac{\partial^2 u}{\partial y_i^2}(x-y) \right| dy + \eta |B_h| \right) \\ &\leq \frac{C}{\epsilon^2} \frac{|A_\epsilon|}{|B_h|} + \eta \\ &\leq \frac{C}{\epsilon^2} \left(\frac{\delta}{\epsilon}\right)^{1/2} \frac{1}{|B_h|} + \eta, \quad x \in \Omega_{2\epsilon}. \end{aligned}$$

Choosing $h = C \left(\frac{1}{\omega_n}\right)^{1/n} \left(\frac{\delta}{\epsilon}\right)^{\frac{1}{4n}} \frac{1}{\epsilon^{2/n}}$ (observe explicitly that $h < \frac{\epsilon}{1+\delta}$ if ϵ is sufficiently large with respect to δ), we get

$$\left| \frac{\partial^2 u_h}{\partial x_i^2}(x) - 1 \right| \leq 2\eta, \quad x \in \Omega_{2\epsilon}. \quad (26)$$

Analogously, if $i \neq j$

$$\left| \frac{\partial^2 u_h}{\partial x_i \partial x_j}(x) \right| \leq \frac{C}{\epsilon^2} \left(\frac{\delta}{\epsilon} \right)^{1/2} \frac{1}{|B_h|} + \eta, \quad x \in \Omega_{2\epsilon},$$

and, from the choice of h , it holds

$$\left| \frac{\partial^2 u_h}{\partial x_i \partial x_j}(x) \right| \leq 2\eta, \quad x \in \Omega_{2\epsilon}. \quad (27)$$

Moreover, from the choice of h and from (24) we can easily deduce

$$\Omega_{4\epsilon} \subseteq \Omega_{3\epsilon, h} \subseteq \Omega_{2\epsilon}, \quad (28)$$

where $\Omega_{3\epsilon, h} = \{u_h < -3\epsilon\}$. Using (26), (27) and (28) we obtain that the matrix $D^2 u_h(x)$ is positive defined in $\Omega_{3\epsilon, h}$; hence, the connected components of $\Omega_{3\epsilon, h}$ are convex. Let us consider a connected component $\Omega_{3\epsilon, h}^i$ of $\Omega_{3\epsilon, h}$ and assume that u_h attains its minimum in $\Omega_{3\epsilon, h}^i$ at the point x_0 . Setting

$$v^i(x) = u_h(x_0) + \frac{1}{2}|x - x_0|^2, \quad x \in \Omega_{3\epsilon, h}^i,$$

and using Taylor expansion, by (26), (27) again, we get

$$\begin{aligned} \|u_h - v^i\|_{L^\infty(\Omega_{3\epsilon, h}^i)} &\leq n\eta d^2; \\ \|Du_h - Dv^i\|_{L^\infty(\Omega_{3\epsilon, h}^i)} &\leq 2n\eta d. \end{aligned} \quad (29)$$

Let $\gamma = 3\epsilon + n\eta d^2$ and let $B^i = \{x: v^i(x) < -\gamma\}$ (see Fig. 3). Then $B^i \subset \Omega_{3\epsilon, h}^i$ and

$$\begin{aligned} \|u - v^i\|_{L^\infty(B^i)} &\leq \|u - u_h\|_{L^\infty(B^i)} + \|u_h - v^i\|_{L^\infty(B^i)} \\ &\leq h(1 + \delta) + n\eta d^2 \\ &\leq h(1 + \delta) + C \left(\frac{\delta}{\epsilon} \right)^{1/4} d^2. \end{aligned}$$

Recalling the value of h and choosing $\epsilon = [C(\frac{1}{\omega_n})^4 \delta(1 + \delta)^{4n}]^{\frac{1}{4n+9}}$, we can write

$$\|u - v^i\|_{L^\infty(B^i)} \leq C \delta^{\frac{1}{4n+9}}. \quad (30)$$

Analogously,

$$\begin{aligned} \|Du - Dv^i\|_{L^\infty(B^i)} &\leq \|Du - Du_h\|_{L^\infty(B^i)} + \|Du_h - Dv^i\|_{L^\infty(B^i)} \\ &\leq (1 + 2\eta)h + 2n\eta d \\ &\leq C \delta^{\frac{1}{4n+9}}. \end{aligned} \quad (31)$$

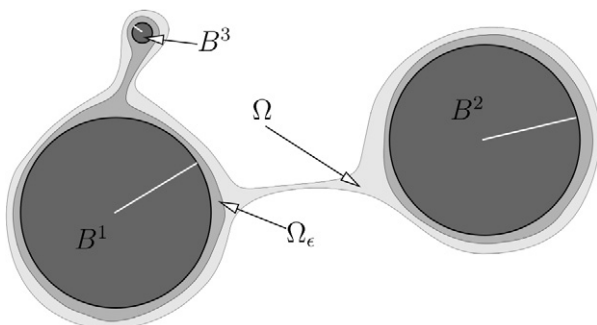


Fig. 3. The figure shows a feasible set Ω , the sublevel set Ω_ϵ and the balls B^i according to Theorem 5.

Adding (30) and (31) we get (21). Moreover, recalling (28), (29) and that $\|u - u_h\|_{L^\infty(\Omega_{2\epsilon})} \leq h(1 + \delta) < \epsilon$, we get

$$\begin{aligned}
 \|u\|_{L^\infty(\Omega \setminus \bigcup_i B^i)} &\leq \max\{\|u\|_{L^\infty(\Omega \setminus \Omega_{2\epsilon})}, \|u\|_{L^\infty(\Omega_{2\epsilon} \setminus \bigcup_i B^i)}\} \\
 &\leq 2\epsilon + \|u\|_{L^\infty(\Omega_{2\epsilon} \setminus \bigcup_i B^i)} \\
 &\leq 2\epsilon + \max\{\|u\|_{L^\infty(\Omega_{2\epsilon} \setminus \Omega_{3\epsilon,h})}, \|u\|_{L^\infty(\Omega_{3\epsilon,h} \setminus \bigcup_i B^i)}\} \\
 &\leq 6\epsilon + \|u - u_h\|_{L^\infty(\Omega_{3\epsilon,h} \setminus \bigcup_i B^i)} + \|u_h\|_{L^\infty(\Omega_{3\epsilon,h} \setminus \bigcup_i B^i)} \\
 &\leq 7\epsilon + \sup_i \|u_h - v^i\|_{L^\infty(\Omega_{3\epsilon,h} \setminus B^i)} + \gamma \\
 &\leq 7\epsilon + \eta d^2 n + \gamma \\
 &\leq C\delta^{\frac{1}{4n+9}}. \quad \square
 \end{aligned}$$

Now we are ready to prove our main theorem.

Proof of Theorem 1. We keep the notation used above for h , u_h , v^i , Ω_σ , $\Omega_{\sigma,h}$, B^i and we first consider the case when $\Omega_{3\epsilon,h}$ is connected. From Theorem 5 we get the existence of a ball $B \subset \Omega$ having radius R and a paraboloid v such that

$$\|u - v\|_{C^1(B)} \leq C\delta^{\frac{1}{4n+9}}, \quad \|u\|_{L^\infty(\Omega \setminus B)} \leq C\delta^{\frac{1}{4n+9}}.$$

Then

$$\begin{aligned}
 \int_{\Omega} (-u) &= \int_B (-v) + \int_B (v - u) + \int_{\Omega \setminus B} (-u) \\
 &\leq \frac{\omega_n R^{n+2}}{n+2} + \|v - u\|_{L^\infty(B)} |\Omega| + \|u\|_{L^\infty(\Omega \setminus B)} |\Omega| + \gamma \omega_n R^n \\
 &\leq \frac{\omega_n R^{n+2}}{n+2} + C\delta^{\frac{1}{4n+9}}.
 \end{aligned} \tag{32}$$

Since $|Du| \leq 1 + \delta$ on the boundary of $\Omega_{3\epsilon}$, then, by (21), on the boundary of B it holds $|Dv| = R \leq 1 + C\delta^{\frac{1}{4n+9}}$. So, from (18) and (20) we can deduce

$$(1 - \delta)^n \frac{\omega_n}{n+2} - C\delta \leq \frac{|\Omega|}{n+2} - C\delta \leq \frac{R^2|B|}{n+2} + C\delta^{\frac{1}{4n+9}} \leq \frac{|B|}{n+2} + C\delta^{\frac{1}{4n+9}},$$

that gives

$$|\Omega \triangle B| \leq C\delta^{\frac{1}{4n+9}} \quad \text{and} \quad R \geq 1 - C\delta.$$

If $\Omega_{3\epsilon, h}$ is not connected, we may have up to a countable set of connected components. Let us denote by R_i the radius of the ball B^i . Not necessarily all the radii R_i are close to one, since

$$\sum_i \omega_n R_i^{n+2} = \sum_i |B^i| R_i^2 \leq (1 + C\delta^{\frac{1}{4n+9}})^2 |\Omega|,$$

where, as before, we have used $R_i \leq 1 + C\delta^{\frac{1}{4n+9}}$. Then, using (21) and (22) as for (32) we have

$$\begin{aligned} \int_{\Omega} (-u) &\leq \sum_i \int_{B^i} (-v^i) + \sup_i \|v^i - u\|_{L^\infty(B^i)} |\Omega| + \|u\|_{L^\infty(\Omega \setminus \bigcup_i B^i)} |\Omega| \\ &\leq \sum_i \frac{\omega_n R_i^{n+2}}{n+2} + C\delta^{\frac{1}{4n+9}}. \end{aligned} \quad (33)$$

(33) and (18) immediately imply that

$$\left| \Omega \triangle \bigcup_i B^i \right| \leq C\delta^{\frac{1}{4n+9}}.$$

On the other hand, from (18)

$$\int_{\Omega} (-u) \geq \frac{|\Omega|}{n+2} - C\delta \geq \sum_i \frac{|B^i|}{n+2} - C\delta;$$

hence

$$\sum_i |B^i| - C\delta \leq \sum_i |B^i| R_i^2 + C\delta^{\frac{1}{4n+9}}.$$

For a fixed k we get

$$\begin{aligned} \omega_n R_k^n + \sum_{i \neq k} |B^i| &\leq \omega_n R_k^{n+2} + \sum_{i \neq k} |B^i| (1 + C\delta^{\frac{1}{4n+9}})^2 + C\delta^{\frac{1}{4n+9}} \\ &\leq \omega_n R_k^{n+2} + \sum_{i \neq k} |B^i| + C\delta^{\frac{1}{4n+9}} |\Omega|. \end{aligned}$$

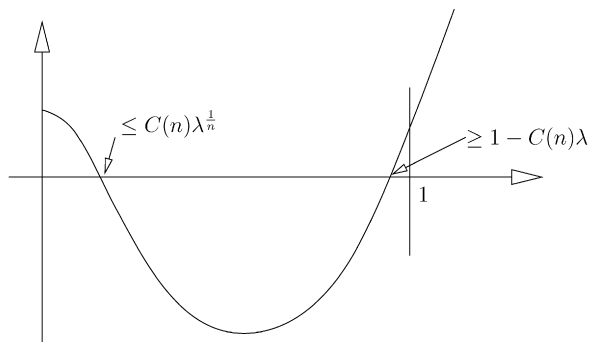


Fig. 4. The schematic behavior of the function $x^{n+2} - x^n + \lambda$.

The last one implies $R_k^{n+2} - R_k^n + \lambda \geq 0$ (see Fig. 4) where $\lambda = C\delta^{\frac{1}{4n+9}} \frac{|\Omega|}{\omega_n}$, therefore $R_k \leq C(n)\lambda^{\frac{1}{n}}$ or $R_k \geq 1 - C(n)\lambda$. Let us denote by B_s^l , those balls having radius $R_l \leq C(n)\lambda^{\frac{1}{n}}$ and by B_g^j , $j = 1, \dots, m$, those ones having radius $R_j \geq 1 - C(n)\lambda$. We want to show that in fact

$$\left| \Omega \triangle \bigcup_{j=1}^m B_g^j \right| \leq C\delta^{\frac{2}{n(4n+9)}}.$$

Indeed, by (33), (18) and the fact that $R_i \leq 1 + C\delta^{\frac{1}{4n+9}}$ we get

$$\begin{aligned} \frac{|\Omega|}{n+2} - C\delta^{\frac{1}{4n+9}} &\leq \sum_i \int_{B^i} (-v^i) + C\delta^{\frac{1}{4n+9}} \\ &= \sum_{j=1}^m \int_{B_g^j} (-v^j) + \sum_l \int_{B_s^l} (-v^l) + C\delta^{\frac{1}{4n+9}} \\ &\leq \sum_{j=1}^m \frac{|B_g^j|}{n+2} + \sum_l \frac{R_l^{n+2} \omega_n}{n+2} + C\delta^{\frac{1}{4n+9}} \\ &\leq \sum_{j=1}^m \frac{|B_g^j|}{n+2} + C\lambda^{\frac{2}{n}} \sum_l \frac{|B_s^l|}{n+2} + C\delta^{\frac{1}{4n+9}} \\ &\leq \sum_{j=1}^m \frac{|B_g^j|}{n+2} + C\lambda^{\frac{2}{n}} \frac{|\Omega|}{n+2} + C\delta^{\frac{1}{4n+9}}, \end{aligned}$$

that implies

$$\frac{|\Omega|}{n+2} \leq \sum_{j=1}^m \frac{|B_g^j|}{n+2} + C\delta^{\frac{2}{n(4n+9)}}, \quad (34)$$

that is

$$\left| \Omega \triangle \bigcup_{j=1}^m B_g^j \right| \leq C \delta^{\frac{2}{n(4n+9)}}.$$

(34) and (20) enforce $m \neq 0$. This together with Theorem 5 conclude the proof. $\square$

4. The smooth case

An a priori prescription on the smoothness of Ω , together with strengthened boundary conditions, can improve the L^1 type estimate (7) obtaining that Ω is almost a unitary ball in a stronger sense. We begin by recalling the definition of $C^{2,\alpha}$ domain (see [5], p. 94).

Definition 6. A bounded open set Ω in $\mathbb{R}^n$ and its boundary are of class $C^{2,\alpha}$, $0 \leq \alpha \leq 1$, if at each point $x_0 \in \partial\Omega$ there is a ball $B = B(x_0)$ and a one-to-one mapping ψ of B onto $D \subset \mathbb{R}^n$ such that

$$\psi(B \cap \Omega) \subset \mathbb{R}_+^n, \quad \psi(B \cap \partial\Omega) \subset \partial\mathbb{R}_+^n, \quad \psi \in C^{2,\alpha}(B), \quad \psi^{-1} \in C^{2,\alpha}(D).$$

Remark 7. From the compactness of $\partial\Omega$ there exist two constants K and ρ_0 such that for all $x \in \partial\Omega$ and $0 < \rho \leq \rho_0$ there exists $x_0 \in \partial\Omega$ and a one-to-one mapping ψ of $B_\rho(x_0)$ onto $D \subset \mathbb{R}^n$ such that $x \in B_\rho(x_0)$ and

$$\begin{aligned} \psi(B_\rho(x_0) \cap \Omega) &\subset \mathbb{R}_+^n; & \psi(B_\rho(x_0) \cap \partial\Omega) &\subset \partial\mathbb{R}_+^n; \\ \|\psi\|_{C^{2,\alpha}(B_\rho(x_0))} &\leq K, & \|\psi^{-1}\|_{C^{2,\alpha}(D)} &\leq K. \end{aligned} \quad (35)$$

The following lemma shows that if u is a smooth solution to problem (13)–(14) in a $C^{2,\alpha}$, bounded, connected, open set, having the gradient bounded away from zero, then its sub-level sets are definitively connected.

Lemma 8. Let Ω be a bounded, connected, open set of class $C^{2,\alpha}$ and let u be a solution to problem (13)–(14) satisfying

$$||Du| - 1| \leq \delta \quad \text{on } \partial\Omega. \quad (36)$$

Then there exists a positive constant $\epsilon_0 = \epsilon_0(n, \alpha, K, \rho_0)$ such that $\Omega_\epsilon = \{u < -\epsilon\}$ is a connected open set for $\epsilon < \epsilon_0$.

Proof. Let us denote

$$\Omega^{\sqrt{2\epsilon}} = \{x \in \Omega: \text{dist}(x, \partial\Omega) \geq \sqrt{2\epsilon}\}$$

and let us firstly prove that $\Omega^{\sqrt{2\epsilon}} \subset \Omega_\epsilon$. Let $\bar{x} \in \Omega^{\sqrt{2\epsilon}}$ and let us consider the function $w(x) = \frac{(x-\bar{x})^2 - 2\epsilon}{2}$. Then $\Delta w = n$ in Ω and $w > 0$ on $\partial\Omega$; by classical maximum principle $u < w$ in Ω , that implies $u(\bar{x}) < w(\bar{x}) = -\epsilon$, i.e. $\bar{x} \in \Omega_\epsilon$.

Now, let $x \in \Omega - \Omega^{\sqrt{2}\epsilon}$ and let $x_0 \in \partial\Omega$ such that $|x - x_0| = \text{dist}(x, \partial\Omega)$. Lagrange theorem says that

$$Du(x) - Du(x_0) = D^2u(\xi)(x - x_0)$$

for some point ξ that belongs to the line segment connecting x and x_0 . From the classical estimates for the second derivatives in smooth domains we deduce that $|D^2u(x)| \leq C(n, \alpha, K, \rho_0)$ at every point $x \in \Omega$. Then

$$||Du(x)| - |Du(x_0)|| \leq C(n, \alpha, K, \rho_0)\sqrt{2}\epsilon$$

and, by condition (36)

$$||Du(x)| - 1| \leq \delta + C(n, \alpha, K, \rho_0)\sqrt{2}\epsilon.$$

Thus, choosing $\epsilon < \frac{1}{2}(\frac{\delta}{C(n, \alpha, K, \rho_0)})^2 = \epsilon_0$ we have

$$||Du(x)| - 1| \leq 2\delta, \quad \text{if } \text{dist}(x, \partial\Omega) < \sqrt{2}\epsilon.$$

This implies that each point of $\partial\Omega_\epsilon$, $\epsilon < \epsilon_0$, has a neighborhood where $\partial\Omega_\epsilon$ is the graph of a $C^{2, \alpha}$ function of $n - 1$ between the coordinates $x_1, \dots, x_n$, and thus Ω_ϵ is connected. $\square$

From Lemma 8 and Theorem 1 it can be easily deduced the following

Corollary 9. *Let Ω be a bounded, connected, open set of class $C^{2, \alpha}$ and let u be a solution to problem (13)–(14) satisfying (36). Then there exist a positive constant $C = C(n, d, \alpha)$ and a ball $B_R \subset \Omega$ such that*

$$\begin{aligned} |1 - R| &\leq C\delta^{\frac{1}{4n+9}}, \\ |\Omega \Delta B_R| &\leq C\delta^{\frac{1}{4n+9}}, \\ |\mathcal{H}^{n-1}(\partial\Omega) - n\omega_n R^n| &\leq C\delta^{\frac{1}{4n+9}}. \end{aligned} \tag{37}$$

We go on by proving the following technical lemma, where we show that, if Ω is close to a ball in symmetric difference and in perimeter, then it may have tentacles, but these tentacles do not count “too much” in terms of perimeter.

Lemma 10. *Let Ω be a bounded, connected, open set in $\mathbb{R}^n$ such that*

$$|\Omega \Delta B_R| < \sigma, \quad |P(\Omega) - n\omega_n R^{n-1}| < \sigma \tag{38}$$

for some ball $B_R = B_R(x)$ and $\sigma > 0$ small enough. Then, there exists $\bar{R} > R$ such that

$$\bar{R} - R < \sigma^{1/2}, \quad P(\Omega \setminus B_{\bar{R}}) - \mathcal{H}^{n-1}(\Omega \cap \partial B_{\bar{R}}) \leq C(n, |\Omega|)\sigma^{1/2}.$$

Proof. By co-area formula

$$\int_R^\infty \mathcal{H}^{n-1}(\Omega \cap \partial B_r) dr = |\Omega \setminus B_R| \leq |\Omega \triangle B_R| < \sigma;$$

then, there exists $\bar{R} > R$ such that

$$\bar{R} - R \leq \sigma^{1/2}, \quad \mathcal{H}^{n-1}(\Omega \cap \partial B_{\bar{R}}) \leq \sigma^{1/2}. \quad (39)$$

On the other hand, by classical isoperimetric inequality and assumption (38)

$$\begin{aligned} P(\Omega \cap B_{\bar{R}}) &\geq n\omega_n^{1/n} |\Omega \cap B_{\bar{R}}|^{1-1/n} \\ &\geq n\omega_n^{1/n} (\omega_n R^n - 2\sigma)^{1-1/n} \\ &\geq (P(\Omega) - \sigma) \left(1 - \frac{2\sigma}{|\Omega| - \sigma}\right)^{1-1/n} \end{aligned}$$

and then, since

$$\begin{aligned} P(\Omega \cap B_{\bar{R}}) &= P(\Omega) - P(\Omega \setminus B_{\bar{R}}) + 2\mathcal{H}^{n-1}(\Omega \cap \partial B_{\bar{R}}) \\ &\geq (P(\Omega) - \sigma) \left(1 - \frac{2\sigma}{|\Omega| - \sigma}\right)^{1-1/n} \end{aligned}$$

from (39) we get

$$\begin{aligned} &P(\Omega \setminus B_{\bar{R}}) - \mathcal{H}^{n-1}(\Omega \cap \partial B_{\bar{R}}) \\ &\leq P(\Omega) \left[1 - \left(1 - \frac{2\sigma}{|\Omega| - \sigma}\right)^{1-1/n}\right] + \mathcal{H}^{n-1}(\Omega \cap \partial B_{\bar{R}}) + \sigma \left(1 - \frac{2\sigma}{|\Omega| - \sigma}\right)^{1-1/n} \\ &\leq P(\Omega) \left[1 - \left(1 - \frac{2\sigma}{|\Omega| - \sigma}\right)^{1-1/n}\right] + \sigma^{1/2} + \sigma \left(1 - \frac{2\sigma}{|\Omega| - \sigma}\right)^{1-1/n} \\ &\leq C(n, |\Omega|) \sigma^{1/2}. \quad \square \end{aligned}$$

We conclude by proving Theorem 2.

Proof of Theorem 2. Let R and $\bar{R}$ be the radii of the balls produced respectively in Corollary 9 and Lemma 10 and let us denote by $\bar{\bar{R}}$ the radius of the smallest ball containing Ω and concentric with B_R . Clearly $R \leq R_{in}$ and $\bar{\bar{R}} \geq R_{out}$, hence, by Corollary 9 and Lemma 10

$$R_{out} - R_{in} \leq \bar{\bar{R}} - R \leq \bar{\bar{R}} - \bar{R} + C\delta^{\frac{1}{2(4n+9)}}.$$

Hence it suffices to prove that

$$\bar{\bar{R}} - \bar{R} \leq C\delta^\mu, \quad \mu > 0.$$

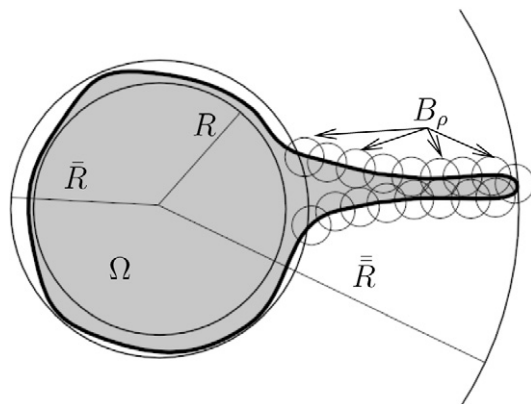


Fig. 5.

Let ρ_0 and K be the constants given by Remark 7; choosing a point $x \in \partial\Omega$ and $\rho = \min\{\rho_0, \frac{\bar{\bar{R}} - \bar{R}}{4}\}$ there exists the mapping ψ satisfying (35).

First of all we observe that $B_{\frac{\rho}{K}}(\psi(x_0)) \subset \psi(B_\rho(x_0))$. Then the application

$$\varphi : y \in B_{\frac{\rho}{K}}(\psi(x_0)) \cap \partial\mathbb{R}_+^n \rightarrow \psi^{-1}(y) \in B_\rho(x_0) \cap \partial\Omega$$

is a parameterization of $B_\rho(x_0) \cap \partial\Omega$ and

$$\mathcal{H}^{n-1}(B_\rho(x_0) \cap \partial\Omega) = \int_{B_{\frac{\rho}{K}}(\psi(x_0)) \cap \partial\mathbb{R}_+^n} \sqrt{\det(D\varphi^T \circ D\varphi)} \geq C(n) \left(\frac{\rho}{K}\right)^{n-1}.$$

Now, since $\partial\Omega$ is compact, we can find a finite set of balls B^j , $j = 1, \dots, l$, of radius ρ such that $\partial\Omega \subset \bigcup_{j=1}^l B^j$ (see Fig. 5). Among them there exist a number q of mutually disjoint balls included in $B_{\bar{\bar{R}}} \setminus B_{\bar{R}}$ (with $q \geq \frac{\bar{\bar{R}} - \bar{R}}{4\rho}$); therefore

$$\mathcal{H}^{n-1}(\partial\Omega \setminus \bar{B}_{\bar{R}}) \geq \frac{\bar{\bar{R}} - \bar{R}}{4\rho} C(n) \left(\frac{\rho}{K}\right)^{n-1} = \frac{\bar{\bar{R}} - \bar{R}}{4} C(n) \left(\frac{\rho}{K}\right)^{n-2}$$

that is

$$\bar{\bar{R}} - \bar{R} \leq C(n, K, \rho_0) \mathcal{H}^{n-1}(\partial\Omega \setminus \bar{B}_{\bar{R}}) \quad \text{if } \rho = \rho_0,$$

$$\bar{\bar{R}} - \bar{R} \leq C(n, K, \rho_0) (\mathcal{H}^{n-1}(\partial\Omega \setminus \bar{B}_{\bar{R}}))^{1/(n-1)} \quad \text{if } \rho = \frac{\bar{\bar{R}} - \bar{R}}{4}.$$

Observing that $\mathcal{H}^{n-1}(\partial\Omega \setminus \bar{B}_{\bar{R}}) = P(\Omega \setminus B_{\bar{R}}) - \mathcal{H}^{n-1}(\Omega \cap \partial B_{\bar{R}})$, by the smoothness of Ω , by Lemma 10 and Corollary 9 we get (8).

Finally, from (8) and (37) it immediately follows that

$$|1 - R_{in}| \leq |1 - R| + |R_{out} - R_{in}| \leq C \delta^{\frac{1}{2(4n+9)(n-1)}}$$

and, analogously,

$$|1 - R_{out}| \leq C \delta^{\frac{1}{2(4n+9)(n-1)}}. \quad \square$$

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